

A data driven approach for reference evapotranspiration forecasting: merging weather forecasts with persistence based Approaches

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ABSTRACT

Data driven models have been extensively used for reference evapotranspiration (ET_o) forecasting using persistence based approaches. Here, a data driven framework is proposed for forecasting ET_o at short and medium range time scales by merging the persistence approach with weather forecasts. Three machine learning (ML) models, namely Support Vector Regression (SVR), Random Forest (RF), and Extreme Gradient Boosting (XGB), and two deep learning (DL) models, namely Long Short Term Memory (LSTM) and Convolutional Neural Network (CNN), are tested within the proposed framework. Weather hindcasts from two Numerical Weather Prediction (NWP) models, namely the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Global Ensemble Forecast System (GEFS), are used in the framework. The framework is then evaluated against forecasts from NWP models, as well as against a conventional NWP bias-correction technique and an existing persistence-based approach commonly used in the literature. Results show that the proposed framework provides better ET_o forecasts than the forecasts from NWP models. Results also show that the performance of the proposed approach is lower in the mountainous regions, owing to the low skill of the NWP forecasts in the region. All ML/DL models show similar skill in forecasting ET_o, with XGB requiring the least time for training and hence is recommended within the framework. Evaluation across crop growing seasons indicates that the proposed approach provides slightly better forecasts for the Rabi season (NRMSE = 3.28% using XGB and ECMWF model) compared to the Kharif season (NRMSE = 3.66% using XGB and ECMWF model), which is due to the low skill of the NWP based forecasts for the Kharif season compared to Rabi. Comparative evaluation also indicates that the proposed framework outperforms both the traditional NWP bias-correction methods as well as the persistence-based approach in literature. The proposed approach can potentially be used for multi-step ahead forecasting of ET_o across the Indian region.

1. Introduction

Reference evapotranspiration (ET_o), an important hydrologic variable, is used for drought characterization, flash drought estimation and estimation of crop water demand (Feng et al., 2019; Mitra et al., 2018; Prabhakar et al., 2023; Varghese et al., 2026; Yang et al., 2021). Forecasting of reference evapotranspiration (ET_o) is therefore important for the overall agricultural water resources management, irrigation scheduling and forecasting of regional water demand (Cai et al., 2011; Perera et al., 2014; Tian and Martinez, 2014; Vanella et al., 2020).

Studies related to forecasting of ET_o have generally employed two approaches (1) persistence-based approach (2) NWP based approach. Traditional forecasting of ET_o has used persistence-based approaches

which involves training statistical/machine learning algorithms with time lagged weather variables (Elzain et al., 2024; Gong et al., 2021; Lucas et al., 2020; Zhao et al., 2019) or ET_o information (Ahmadi et al., 2023; Ferreira and Da Cunha, 2020; Karbasi et al., 2022). For instance, Nourani et al. (2020) used lagged meteorological data such as temperature, relative humidity, solar radiation for multi-step ahead forecasting of ET_o using classical machine learning (ML) algorithms (Nourani et al., 2020). One of the shortcomings of the traditional forecasting techniques (based on persistence) is that they only consider time lagged variables and do not consider future meteorological information. Nevertheless, persistence-based techniques employing ML models remain popular for ET_o forecasting, primarily due to the limited availability of the full range of variables required for ET_o estimation from public weather forecasts.

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The NWP based approach entails leveraging the recent advances in numerical weather predictions (NWP) by forecasting ETo with variables derived from forecasts of NWP models (Karbasi et al., 2022; Liu et al., 2020; Zhang et al., 2023). Yang et al. (2019) used forecasted weather variables from public weather forecasts within the FAO Penman Monteith (PM) formulation for forecasting of ETo at short to medium range time scales (Yang et al., 2019). Similarly, Tian and Martinez (2014) used forecast variables from the Global Ensemble Forecast System (GEFS) for forecasting of ETo (Tian and Martinez, 2014). However, NWP model forecast variables contain biases that are also reflected in the ETo forecasts estimated (using the PM method) and hence cannot be directly used for decision making purposes (Saminathan et al., 2024). Therefore, various statistical techniques such as Bayesian model averages and Non-homogenous gaussian regression have been proposed for post-processing NWP based ETo forecasts (Medina et al., 2018; Tian and Martinez, 2012).

A recent line of research on NWP-based approaches has employed only temperature forecasts from NWP models in combination with data-driven (ML/DL) models to predict ETo (Luo et al., 2015; Zhang et al., 2023). For example, Zhang et al. (2023) used temperature forecasts from public weather forecast information for training ML/DL models for forecasting of ETo in China. However, these approaches rely heavily on the ML/DL model's ability to capture the complex non-linear relationships inherent in the FAO-Penman-Monteith (PM) method using temperature data alone (Alves et al., 2017; Reis et al., 2019; Zhang et al., 2022). Additionally, the models must correct for biases present in the NWP-derived temperature forecasts. Since accurate ETo estimation via the PM method requires multiple meteorological variables, relying solely on temperature forecasts may reduce predictive skill. To address this limitation, the present study proposes a novel approach that integrates persistence-based data with NWP forecasts within an ML/DL framework for enhanced ETo forecasting. The proposed approach involves initially estimating short- to medium-range (1–7 days) ETo forecasts from the weather variables provided by NWP models using the PM formulation. These forecasts are subsequently combined with time-lagged ETo estimates to generate predictions through ML/DL algorithms. This integration helps correct biases in the NWP model outputs by leveraging observed time-lagged ETo values (via a persistence-based approach) and training the ML/DL algorithms using both the observed ETo and NWP-based forecasts, resulting in improved ETo predictions. To the best of our knowledge no studies have merged the persistence-based and NWP based forecasts within ML/DL framework for forecasting of ETo. Further, comparative assessment of using NWP based forecasts against the plethora of persistence based approaches in literature are severely lacking. Therefore, in this study we also do a comparative assessment of the proposed approach against NWP based forecasts, and persistence based approaches. This will, for the first time, allow quantification of the improvement in ETo forecast performance achieved by using NWP information compared to previous approaches. Further, the proposed approach is compared with one of the most widely used NWP bias-correction methods to evaluate its efficacy.

India is a large agricultural nation with extensive agricultural water requirements. The need for irrigation water strains surface and groundwater supplies, endangering their sustainability (Gedam et al., 2023; Roy et al., 2021). Current irrigation practices in the region do not incorporate forecasts, partly due to lack of information on meteorological forecasts. There is an urgent need for including forecast information for irrigation optimization at regional and local levels towards meeting sustainability goals. Some studies recently have reported on ETo forecasts in India, however have mainly been based on persistence approaches and do not incorporate NWP information (Mandal and Chanda, 2023; Patil and Deka, 2017; Rajput et al., 2023; Saggi and Jain, 2019; Sharma et al., 2022). However, none of the studies on ETo forecasting in India have incorporated NWP based forecast information. Therefore, in this study we assess the performance of the proposed methodology for forecasting of ETo across different climate zones and

agricultural seasons in India. We also compare the efficacy of the proposed methodology against the NWP based, traditional NWP bias-correction approach and the persistence-based approach (Mandal and Chanda, 2023). The overall goal of this study is to develop a novel framework for forecasting ETo at short to medium range time scales by merging persistence-based and NWP based forecasting approaches within ML/DL framework. The study also has the following objectives (i) Evaluation of the different ML/DL models for ETo forecasting within the proposed approach (ii) Assessment of the forecasts from the proposed ML/DL based framework towards addressing the biases in the NWP forecasts (iii) Comparison of the proposed approach against persistence-based forecasting approach.

2. Materials and methods

2.1. Study area

The proposed framework was tested over the different climate zones across the Indian region (Fig. 1). The Köppen–Geiger climate classification, the most widely used classification, is used in this study for climate zone classification (Chen and Chen, 2013; Yaduvanshi et al., 2021). The study region is classified into arid (Z1), mountainous (Z2), tropical (Z3), and temperate (Z4) climate zones. The framework was also tested across two major agricultural seasons in India namely Kharif and Rabi. The Kharif season extends from June to September during the Indian monsoon period, while Rabi is the winter cropping season that lasts from October to March (BIRTHAL and Hazrana, 2019). The analysis was carried at $1^\circ \times 1^\circ$ grid point spacing over the entire study area.

2.2. Data used

The meteorological variables required for ETo estimation using the PM method include temperature maximum (Tmax) and minimum (Tmin), windspeed (Ws), solar radiation (Rs), and relative humidity (RH). The above mentioned variables were procured from India Meteorological Department (IMD) (<https://cdsp.imdpune.gov.in/>) and ERA5 reanalysis dataset for the entire study area. Reanalysis datasets are considered as a proxy for observations and have been extensively used in ETo forecasting studies (Mandal and Chanda, 2023; Purnadurga et al., 2019; Saggi and Jain, 2019; Tarek et al., 2020; Varghese and Mitra, 2025; Veysi et al., 2024). Previous research has also highlighted the reliability of ERA5 and its suitability as a substitute or complement to observational data, particularly in regions with sparse station coverage (Hassler and Lauer, 2021; Huang et al., 2016; Muñoz-Sabater et al., 2021; Saini and Attada, 2023; Vishal and Rani, 2022). Here the ERA5 reanalysis dataset is used due to the unavailability of weather variables from ground based stations within the study region. Even though a few stations exist, but the duration of the datasets vary, hindering comprehensive analysis and comparison across different regions. Further, recent studies by Veysi et al. (2024) and Tarek et al. (2020) also reported that the ERA5 reanalysis dataset showed strong correlation with the ground measurements. Over the Indian subcontinent, ERA5-derived variables have been widely used for reference evapotranspiration (ETo) forecasting (Aadhar and Mishra, 2020; Mahto and Mishra, 2019; Mandal and Chanda, 2023; Varghese and Mitra, 2024).

The forecasted ETo was obtained from weather variables of two different NWP models, namely the European Centre for Medium Range Weather Forecasts (ECMWF) and Global Ensemble Forecast System (GEFS). The ECMWF forecasts have 50 ensemble members, and it was downloaded from the THORPEX Interactive Grand Global Ensemble (TIGGE) archive: <http://apps.ecmwf.int/datasets/data/tigge>. The ECMWF data from the years 2007–2019 was used for ETo estimation. The GEFS forecast system consists of 11 ensemble members; however, only five ensembles were used in this study because only five members are available in hindcast mode which was downloaded from the National Oceanic and Atmospheric Administration website. The GEFS data

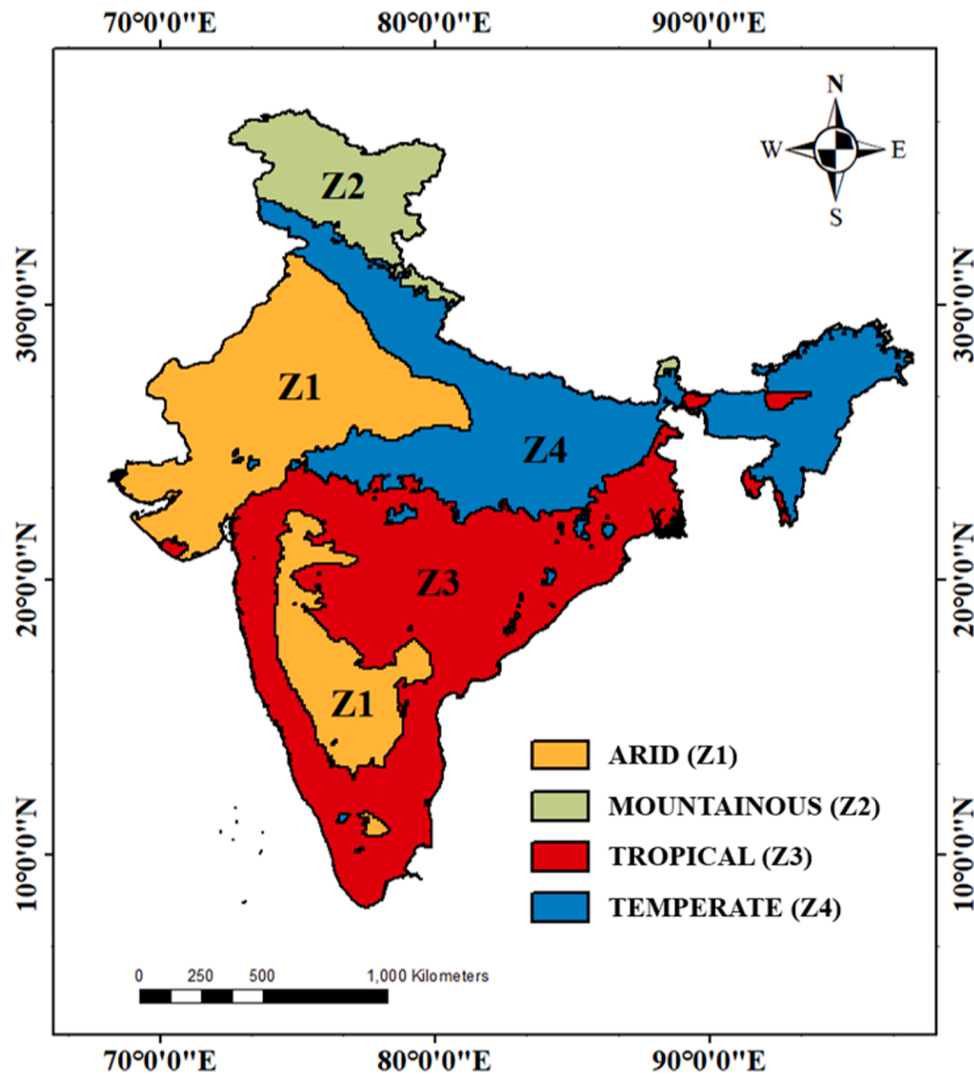


Fig. 1. Different climatic zones considered in the study area based on Köppen-Geiger climate classification.

available from the years 2000–2019 was used for ETo estimation. More details about the dataset can be found in Saminathan et al., 2021 and 2024 (Saminathan et al., 2024, 2021). The time resolution of the data is daily. The estimated ETo from NWP models of the lead 1 and lead 7 day is used in this study.

2.3. ETo estimation

In this study, ETo is estimated using the PM equation. The PM method is proven to be the most accurate and reliable method for ETo estimation compared to other existing methods (Almorox et al., 2018; Cai et al., 2007; Li et al., 2022; Lin et al., 2023; Yang et al., 2019, 2016; Zhang et al., 2022). It is recommended by the Food and Agricultural Organization (FAO). The PM equation is given by

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \left(\frac{900}{T + 273} \right) u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

In equation 1, ETo is the reference evapotranspiration in mm/day; R_n and G are the net radiation at the crop surface and soil heat flux density in $\text{MJ}/\text{m}^2/\text{day}$. T and u_2 are the daily mean air temperature in $^{\circ}\text{C}$ and wind speed in m/s . Δ is the slope of the vapor pressure versus temperature curve in $\text{kPa}/^{\circ}\text{C}$; γ is the psychrometric constant; e_s and e_a is the saturation and actual vapor pressure in kPa .

Equation 1 was used for estimation of observed (from India Meteorological Department, ERA5) and forecasted (from ECMWF and GEFS) ETo at one and seven days lead for the year 2000 to 2019. In the computation of observed ETo the temperature data was taken from IMD as IMD is based on ground stations and would be superior to the temperature data from ERA5. However, due to the lack of other variables at scale across the nation from IMD, corresponding variables from ERA5 were used. Other studies have also used combination of ERA5 and IMD data for ETo studies (Purnadurga et al., 2019). This observed and forecasted ETo was used as input in the framework.

3. Methodology

3.1. Proposed framework

The proposed framework integrates NWP-based forecasts with a persistence-based approach using machine learning (ML) and deep learning (DL) techniques. Specifically, three categories of predictors were employed to forecast ETo at a given lead day: (1) NWP-derived ETo forecasts corresponding to that lead day, (2) Julian day information of the prediction date, and (3) time-lagged ETo estimates. The seven-day lagged ETo values from ERA5 data were utilized to capture persistence effects within the ML/DL models. Julian day information was included to represent seasonal variations in the forecasts. For instance, if

the ETo forecast is made on 20 January (t day) for a one-day lead (t+1 day; 21 January), the predictors would include: the one-day lead NWP forecast for 21 January (t+1 day), the Julian day of 20 January (t day), and time-lagged observed ETo from the previous seven days (t to t-6 days). The target variable (predictand) is the observed ETo for 21 January (t+1 day). Incorporating both lagged ETo values and NWP-based forecasts enables the model to account for physical model information and persistence effects, thereby improving forecast accuracy. The proposed framework was implemented at individual grid points to generate forecasts at 1-day and 7-day lead times. Model performance was assessed across various climatic zones and major cropping seasons within the study area. Three ML technique: Support Vector Regression (SVR), Random Forest (RF), and Extreme Gradient Boosting (XGB) and two DL techniques: Long Short-Term Memory (LSTM) and Convolutional Neural Network (CNN) were evaluated to determine their forecasting skill and computational efficiency (see Fig. 2). Details regarding these ML/DL techniques are provided in the supplementary material. The predictors and predictand information as mentioned above are used as an input to ML/DL techniques and are provided in the Table 1.

Subsequently, the performance of the proposed framework was evaluated against three different setups: (i) NWP-based forecasts, (ii) a widely used bias-correction method for NWP models, and (iii) a persistence-based approach. The first setup compares the proposed approach with the raw NWP based forecasts itself. Thereafter, in the second setup the proposed framework also was compared against an established NWP forecast bias-correction method to evaluate if the proposed method provides better forecasts than ones already established in literature. A wide range of statistical bias-correction approaches have been developed over the years to improve NWP-based forecasts by integrating them with observations. Techniques such as Bayesian Model Averaging (BMA) and Non-Homogeneous Gaussian Regression have been extensively applied to enhance forecasts of meteorological variables including precipitation, temperature, and, in some cases, ETo (Ahmed et al., 2020; Barlow et al., 2018; Hernández et al., 2011; Ji et al., 2019; Samal et al., 2023; Wilson et al., 2007;). In this study therefore the BMA method was compared against the proposed method. Details regarding the BMA approach is provided in the supplementary material. Further the proposed approach was compared against the persistence-based approach outlined by Mandal and Chanda (2023). In the persistence-based approach, only time-lagged ETo values and Julian day information were considered as predictors. The XGB model was exclusively employed for this approach, as it demonstrated high accuracy compared to other ML/DL models while requiring significantly less

computational time.

3.2. Hyperparameter tuning and K-Fold cross validation

As the inputs to the ML/DL technique are extracted, the next step is hyperparameter optimization/tuning in each ML technique. The details of the hyperparameter used are shown in Table S1. The ML/DL techniques were tested using k-fold validation as shown in Figure S1 of supplementary materials at each grid point. Here, “K” is taken as 4, and the entire dataset is divided into four folds. The hyperparameter was tuned using the leave-out “K” fold validation. For instance, while testing the first fold, remaining folds was used for the hyperparameter tuning and the best-performing hyperparameter was tested over the first fold. The average skill from all the k folds at each grid location is considered as the overall skill of the respective ML/DL approach.

3.3. Framework assessment

The proposed ML/DL framework for lead 1 and 7 day ETo forecasting is validated against the observed ETo using five different evaluation metrics namely, the normalized root mean square error (NRMSE), mean absolute error (MAE), root mean square Error (RMSE), coefficient of determination (R^2) and bias (BIAS). The numerical formulations of the metrics are as follows:

$$\text{NRMSE} = \frac{\sqrt{\frac{\sum_{i=1}^p |V_i - M_i|^2}{p}}}{\bar{V}} \times 100\% \quad (2)$$

$$\text{MAE} = \frac{\sum_{i=1}^p |V_i - M_i|}{p} \quad (3)$$

$$R^2 = \frac{(\sum_{i=1}^p [V_i - \bar{V}][M_i - \bar{M}])^2}{(\sum_{i=1}^p [V_i - \bar{V}]^2)(\sum_{i=1}^p [M_i - \bar{M}]^2)} \quad (4)$$

$$\text{BIAS} = \frac{\sum_{i=1}^p V_i - M_i}{\sum_{i=1}^p V_i} \quad (5)$$

In the above equation, “V” and “M” represent the observed and forecasted ETo. “p” represents the total number of days. The NRMSE ranges from 0% to 100%. The MAE ranges from 0 to ∞ ; the lower the NRMSE and MAE, the better the model's performance. R^2 measures the proportion of the variance in the observed values that is explained by the forecasted values, with a higher R^2 indicating a better fit of the forecast

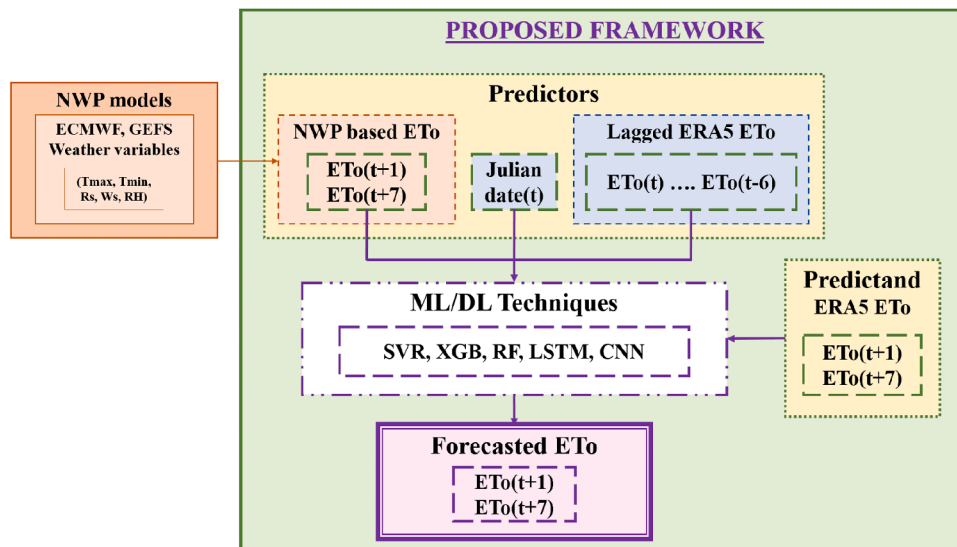


Fig. 2. Flow chart of the proposed framework and overall methodology.

Table 1

Predictand and predictors used in ML/DL models for ETo forecast estimation at different forecast leads.

Forecast lead	ML/DL target variable/ predictand	Predictors	Predictor / input variables calculation
Lead 1	Observed ERA5 ETo (t+1) (PM)	$ETo \sim \left\{ \begin{array}{l} NWP ETo(t+1), \\ \text{lagged observed } ETo(t \dots t-6), \\ \text{Julian date}(t) \end{array} \right\}$	$NWP ETo \sim \left\{ \begin{array}{l} NWP T_{max}, \\ NWP T_{min}, \\ NWP \text{ Relative humidity}, \\ NWP \text{ Solar radiation}, \\ NWP \text{ Wind speed} \end{array} \right\}$
Lead 7	Observed ERA5 ETo (t+7) (PM)	$ETo \sim \left\{ \begin{array}{l} NWP ETo(t+7), \\ \text{lagged observed } ETo(t \dots t-6), \\ \text{Julian date}(t) \end{array} \right\}$	$observed ETo \sim \left\{ \begin{array}{l} IMD T_{max}, \\ IMD T_{min}, \\ ERA5 \text{ Relative humidity}, \\ ERA5 \text{ Solar radiation}, \\ ERA5 \text{ Wind speed} \end{array} \right\}$

*t – refers to the time step

to the observed data. The BIAS value illustrates the consistent difference between forecasted and observed values, with positive bias indicating an overestimation tendency and negative bias suggesting an underestimation tendency of the forecast.

4. Results

The performance of the five ML/DL techniques is evaluated within the proposed framework. In the results section we also present a comparative analysis of the proposed framework against the NWP forecasts, a traditional bias-correction approach and an existing persistence based ML/DL approach.

4.1. Forecast performance of the proposed framework against NWP forecasts and comparison across ML/DL approaches

The domain averaged ETo forecast performance of the NWP based forecasts across lead 1 and 7 days are shown in Table 3, Table S2 and Figure 3. The domain averaged NRMSE and MAE for the ECMWF ETo forecasts are 5.38% and 0.64 mm/d, respectively at lead 1, while the same was 7.62 % and 0.97 mm/d, respectively for the GEFS model (Table 3). Based on the NRMSE and MAE values it is evident that the ECMWF forecasts are better than the GEFS forecasts over the Indian region. Further, it is also evident that the lead 1 forecasts are better than the lead 7 forecasts (Fig. 3). The NWP forecasts show relatively good R^2 values of 0.83 and 0.75 for the ECMWF and GEFS models, respectively at lead 1. The BIAS of the forecasts from the ECMWF model was found to be low (0.05 mm/d), while the GEFS model exhibited moderate BIAS (0.46 mm/d; Table 3).

Table 3 and Table S2 also shows the performance of the five ML/DL approaches within the proposed framework. It is evident that the ML models within the proposed framework are able to considerably improve the performance of the forecasts from both the NWP models. All the ML approaches show very similar performance. For instance, the NRMSE of SVR, RF and XGB are 2.92%, 2.96% and 2.93%, respectively for the ECMWF model at lead 1. The MAE of the ML approaches are considerably lower than NWP based forecasts with domain averaged values equal to 0.35 mm/d and 0.39 mm/d for the ECMWF and GEFS, respectively (for all the three ML techniques). The proposed approach also substantially improves the R^2 values, reaching 0.91 and 0.88 (for the RF model) for ECMWF and GEFS, respectively. In addition, it considerably reduces the BIAS, particularly for the GEFS model, where the NWP-based ETo forecast BIAS was 0.46 mm/day, while the BIAS from the RF-based framework was nearly negligible (−0.01 mm/day; Table 3). The proposed framework achieves a much larger improvement in forecast skill at lead 7 compared to lead 1, although the absolute skill at lead 7 remains lower than that of lead 1. For instance, in the ECMWF model, the change in NRMSE is greater for lead 7 than for lead 1 (Fig. 3). However, the absolute NRMSE values remain higher for lead 7 (ranging from 3.96 to 4.02% for the ML models) compared to lead 1 (ranging from 2.92 to 2.96% for the ML models). Similar results are observed for MAE as well (Fig. 3 and Table 3). The R^2 is lower at lead 7 for ECMWF

model (compared to lead 1) while similar R^2 is observed for the GEFS model (for both lead 1 and 7). In contrast, the BIAS remains comparable for ECMWF and GEFS models across lead 1 and lead 7. Overall, these results indicate that the proposed framework provides substantially improved forecast skill compared to the forecasts generated by the NWP models.

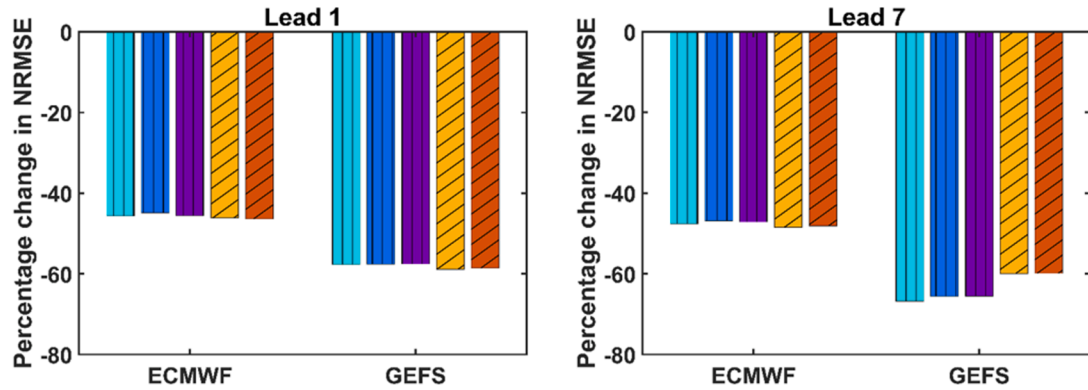
Comparison of the different ML techniques, it was observed that all the ML techniques performed similarly, and no particular technique was found to outperform. For example at lead 1 the range of NRMSE varies between 2.92 to 2.96 % while the MAE values are the same (0.35 mm/d) for the different ML techniques for the ECMWF model. Comparison of the DL models with ML suggests comparable performance at lead 1 (Fig. 3) for both the NWP models. For lead 7 the DL models perform poorer than the ML models for GEFS, while the performance is comparable for the ECMWF (Fig. 3). Since, the performance of the ML/DL models were very similar, we decided to see the time taken for the different ML/DL approaches in the framework. Overall, XGB was found to be taking lesser time (14 minutes), followed by SVR (48 minutes), CNN (59 minutes), RF (77 minutes) and LSTM (116) (Table 2). As the DL models did not show any significant difference in performance, it is suggested to use an ML model, specifically XGB, within the proposed framework. Even though XGB was not the best model based on the skill scores, but the difference in skill score between the best performing model and XGB are negligible (Fig. 3, Table 3, Table S2). Hence, we recommend the XGB model for the framework.

Comparison of the proposed framework across the two NWP models highlight that the performance enhancement using the ML/DL techniques are higher for the GEFS model compared to ECMWF at both the lead times. For instance, reduction in NRMSE, MAE and R^2 are considerably higher for the GEFS model than the ECMWF (Fig. 3). However, starkly different absolute skill scores are achieved using the framework for the two NWP models at lead 1 and lead 7. As mentioned earlier, the ECMWF model forecasts are better than GEFS for both the lead times, and as expected, at lead 1, the forecast skill of the proposed ML/DL based approach is lower for the GEFS model (e.g. NRMSE of 3.23 % using XGB) compared to the ECMWF model (e.g. NRMSE 2.96% using XGB). However, at lead 7 it is observed the skill scores of the ML/DL based approach for GEFS (e.g. NRMSE 3.30 % using XGB) are considerably better than ECMWF (e.g. NRMSE 4.00 % using XGB). Few studies have also reported earlier that statistical techniques are able to considerably improve the skill of GEFS models, particularly at longer lead times, sometimes surpassing ECMWF forecasts in specific scenarios (Hagedorn et al., 2012, 2008; Hamill, 2012; Hamill et al., 2008).

4.2. Comparison of the proposed framework against BMA bias correction approach

From the results of Table 3 it was clear that the BMA improves forecast performance compared to the standalone NWP based forecasts (Table 3 and Table S2). For instance, for the ECMWF forecasts, BMA reduces the NRMSE from 5.38% to 3.63% and MAE from 0.64 to 0.45 mm/day across the entire Indian region (DA), while increasing R^2 from

(a) NRMSE



(b) MAE

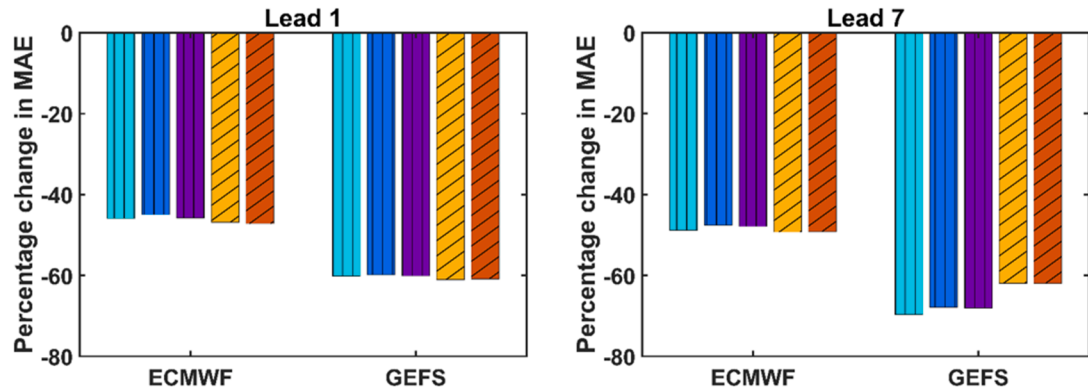
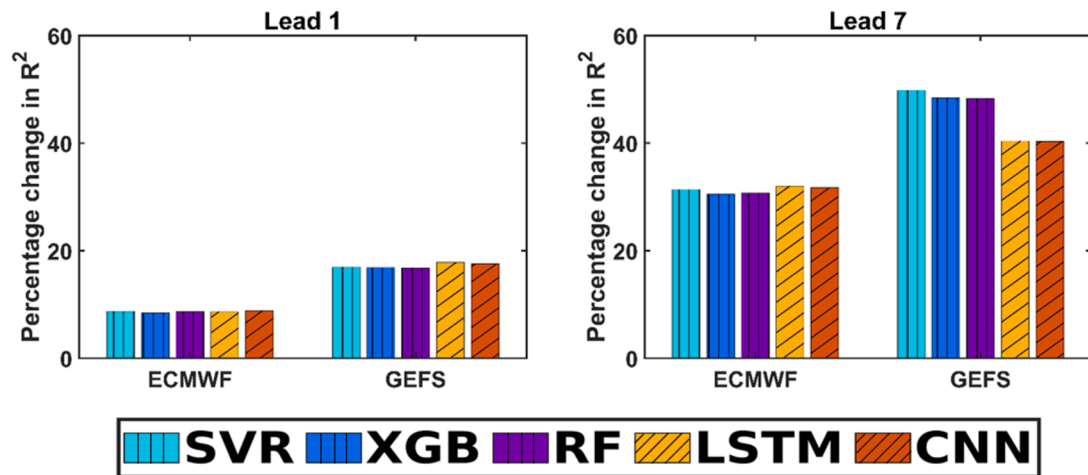
(c) R^2 

Fig. 3. Domain averaged change (proposed approach minus NWP based forecasts) in (a) NRMSE, (b) MAE, and (c) R^2 skill scores on using different techniques across NWP-based ETo over the Indian region for both NWP models at 1 and 7 lead days.

Table 2

Average computational time of the ML/DL approaches at individual grid scale.

Techniques	ML			DL	
	SVR	XGB	RF	LSTM	CNN
Time taken to run for one location (minutes)	48	14	77	116	59
Increase in time with respect to XGB (minutes)	34	-	63	102	45

0.83 to 0.91 (Table 3 and Table S2). Similar improvements are observed in the GEFS forecasts, where NRMSE decreases from 7.62% to 4.68% and MAE from 0.97 to 0.59 mm/day (Table 3 and Table S2). The BIAS values are also substantially reduced, indicating that BMA effectively corrects systematic errors in the ensemble forecasts.

Despite these improvements, the proposed framework consistently outperforms BMA in almost all aspects. For instance, in the ECMWF forecast, the proposed framework achieves NRMSE values around 2.88–2.96% and MAE values around 0.34–0.35 mm/day across the entire Indian region, which are lower than those obtained using BMA.

Table 3

Skill scores of ETo forecasts from NWP models, BMA and the proposed framework (using different ML and DL models) at 1 lead day for the entire study area (domain averaged (DA)) and across different climatic zones (Z1, Z2, Z3, Z4).

Techniques		ECMWF							GEFS						
		NWP based		Proposed framework					NWP based		Proposed Framework				
Metrics	Zones			SVR	XGB	RF	LSTM	CNN			SVR	XGB	RF	LSTM	CNN
NRMSE (%)	DA	5.38	3.63	2.92	2.96	2.93	2.90	2.88	7.62	4.68	3.22	3.23	3.24	3.14	3.16
	Z1	3.49	2.77	2.26	2.32	2.28	2.25	2.20	6.83	3.80	2.68	2.69	2.70	2.53	2.56
	Z2	4.21	4.94	4.09	4.00	3.97	3.86	3.79	7.41	4.68	3.28	3.28	3.23	3.26	3.22
	Z3	3.53	3.38	2.68	2.71	2.69	2.73	2.73	6.46	4.45	2.96	2.97	2.97	2.93	2.95
	Z4	3.44	4.59	3.68	3.75	3.70	3.65	3.66	7.03	5.96	4.21	4.21	4.23	4.08	4.10
MAE (mm/day)	DA	0.64	0.45	0.35	0.35	0.35	0.34	0.34	0.97	0.59	0.39	0.39	0.39	0.38	0.38
	Z1	0.48	0.48	0.37	0.39	0.38	0.37	0.36	0.85	0.68	0.46	0.46	0.46	0.43	0.43
	Z2	0.51	0.42	0.33	0.33	0.32	0.29	0.29	0.94	0.39	0.23	0.24	0.23	0.25	0.24
	Z3	0.50	0.44	0.34	0.34	0.34	0.34	0.34	0.80	0.58	0.37	0.37	0.37	0.37	0.37
	Z4	0.46	0.43	0.34	0.35	0.34	0.33	0.33	0.89	0.58	0.39	0.39	0.39	0.38	0.38
R ²	DA	0.83	0.91	0.91	0.90	0.91	0.91	0.91	0.75	0.84	0.88	0.88	0.88	0.89	0.88
	Z1	0.88	0.93	0.92	0.92	0.92	0.92	0.93	0.76	0.86	0.89	0.89	0.89	0.90	0.90
	Z2	0.88	0.92	0.92	0.92	0.92	0.92	0.93	0.78	0.93	0.94	0.94	0.94	0.94	0.95
	Z3	0.85	0.90	0.90	0.90	0.90	0.89	0.89	0.75	0.82	0.87	0.87	0.87	0.88	0.87
	Z4	0.89	0.89	0.89	0.89	0.89	0.89	0.89	0.77	0.82	0.86	0.86	0.86	0.87	0.87
BIAS (mm/day)	DA	0.05	0	0.02	0	0	0.01	0	0.46	0	0.02	0	-0.01	0	-0.01
	Z1	0.34	0.00	0.02	0	0	0	0	0.25	-0.01	0.02	0	-0.01	-0.01	-0.01
	Z2	0.15	0.03	0.03	0	-0.01	-0.01	0	0.43	0.03	0.01	0	-0.01	-0.01	0
	Z3	0.38	0	0.02	0	0	0.01	0.01	0.16	-0.02	0.02	0	-0.01	0	-0.01
	Z4	0.31	-0.02	0.01	0	0	0.01	0	0.30	0.01	0.02	0	-0.01	0	0

Similarly, for GEFS forecasts, the NRMSE for domain averaged BMA approach ranged between 3.14–3.24% for the proposed approach that is considerably lower than the BMA technique. The R² and BIAS remain similar for BMA and the proposed approach. It is important to note that the proposed approach is able to provide enhanced forecasts in the Z2 region as well (compared to BMA) where the forecast performance from NWP models is particularly low. Overall, based on NRMSE and MAE, the proposed framework is found to be better than the BMA and NWP based forecast.

4.3. Proposed framework performance across climate zones and agricultural cropping seasons

The performance of the proposed ML/DL framework across different

climatic zones and two major agricultural seasons (kharif and rabi) is reported here (Table 3, Table S2, Table 4, Fig. 4, Fig. S2, Fig. S3). Based on NRMSE, MAE, R² and BIAS, zonal evaluation reveals that the Z2 region NWP ETo forecasts are less skillful compared to the rest of the zones. For instance, the NRMSE values of Z2 region for the ECMWF (GEFS) model at lead 1 is 4.21% (7.41%), lower than the rest of the zones (Table 3). Similarly, for the Z2 region, the MAE of NWP forecasts were higher for both the NWP models (compared to rest of the zones). Consequently, we also note that the forecasts from the ML/DL based framework are also poorer for the Z2 region. For instance, for the Z2 region the NRMSE of the RF algorithm is 3.97 %, while for Z1, Z3 and Z4 the values are 2.28 %, 2.69 % and 3.70 %, respectively for ECMWF. Similarly the BIAS of the Z2 was -0.01 while for Z1, Z3 and Z4 the values are 0 using RF algorithm for ECMWF.

Table 4

Skill scores of ETo forecasts from NWP models and the proposed framework (using different ML and DL models) at 1 and 7 lead days for the major agricultural seasons.

Lead 1															
Techniques		ECMWF							GEFS						
		NWP based		Proposed Framework					NWP based		Proposed Framework				
Metrics	Season			SVR	XGB	RF	LSTM	CNN			SVR	XGB	RF	LSTM	CNN
NRMSE (%)	KHARIF	5.12	3.61	3.66	3.62	3.59	3.59	3.59	10.15	3.89	3.92	3.89	3.95	3.98	3.98
	RABI	4.38	3.27	3.28	3.21	3.19	3.14	3.14	6.81	3.56	3.55	3.57	3.34	3.34	3.34
MAE (mm/day)	KHARIF	0.60	0.44	0.45	0.44	0.43	0.43	0.43	0.89	0.48	0.48	0.48	0.49	0.49	0.49
	RABI	0.58	0.25	0.26	0.25	0.26	0.25	0.25	0.94	0.29	0.29	0.29	0.28	0.28	0.28
R ²	KHARIF	0.77	0.82	0.81	0.81	0.82	0.82	0.82	0.68	0.78	0.78	0.78	0.77	0.77	0.77
	RABI	0.81	0.90	0.89	0.90	0.89	0.90	0.90	0.72	0.85	0.85	0.85	0.87	0.87	0.87
BIAS (mm/day)	KHARIF	0.43	0.06	0.02	0.03	0.01	0.02	0.02	0.44	0.06	0.01	0.02	0	0.01	0.01
	RABI	0.04	0	0	-0.01	0	0	0	0.48	0	-0.01	-0.02	-0.01	-0.01	-0.01
Lead 7															
Techniques		ECMWF							GEFS						
		NWP based		Proposed Framework					NWP based		Proposed Framework				
Metrics	Season			SVR	XGB	RF	LSTM	CNN			SVR	XGB	RF	LSTM	CNN
NRMSE (%)	KHARIF	8.43	4.99	5.06	5.04	4.88	4.91	4.91	9.87	3.84	4.00	3.98	4.73	4.74	4.74
	RABI	5.91	4.24	4.32	4.28	4.17	4.22	4.22	7.70	3.52	3.63	3.65	4.17	4.17	4.17
MAE (mm/day)	KHARIF	0.66	0.61	0.63	0.62	0.60	0.60	0.60	0.91	0.46	0.49	0.48	0.59	0.59	0.59
	RABI	0.76	0.34	0.35	0.34	0.34	0.34	0.34	1.03	0.28	0.29	0.30	0.34	0.34	0.34
R ²	KHARIF	0.67	0.66	0.65	0.65	0.67	0.67	0.67	0.62	0.79	0.78	0.78	0.68	0.68	0.68
	RABI	0.63	0.81	0.80	0.80	0.80	0.80	0.80	0.58	0.85	0.84	0.84	0.79	0.79	0.79
BIAS (mm/day)	KHARIF	0.15	0.06	0.01	0.01	0	0.01	0.01	0.46	0.06	0	0.01	0	0.01	0.01
	RABI	0.09	0.01	0.01	0	0	0.01	0.01	0.54	0.01	0	-0.01	0	0	0

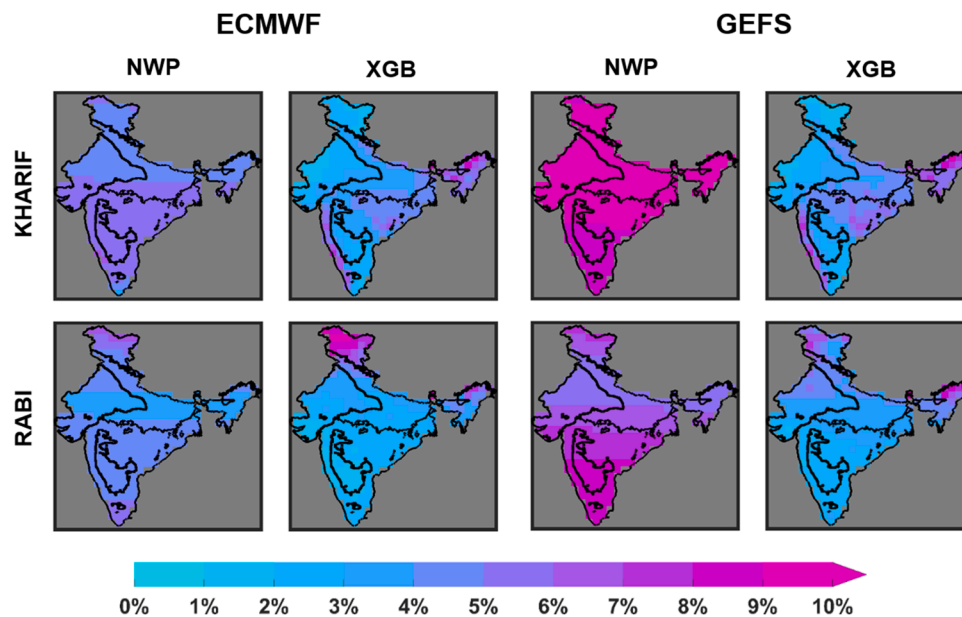


Fig. 4. Gridwise NRMSE values over the Indian region for ETo forecasts estimated using NWP models and the proposed ML/DL (using XGB only) based framework at 1 lead day for the Rabi and Kharif seasons.

Figures 4, S2 and S3 show the gridwise NRMSE and MAE values, respectively derived for NWP forecasts, and the proposed framework for the Rabi and Kharif seasons. Overall it is observed that the NWP ETo forecasts from both the models showed a better skill during the Rabi season compared to Kharif across the study area (Table 4, Fig. 4, Fig. S2, Fig. S3). From Table 4 it is evident that with the proposed framework the skill of Rabi season ETo forecasts are better compared to Kharif forecasts as the NRMSE, MAE, R^2 and BIAS for the Kharif season are higher compared to Rabi season. During the Kharif season, the proposed framework performs comparatively poorly in the northeast and western ghats region with NRMSE higher than or equal to NWP forecasts at some locations (Fig. 4). For the rest of the regions, considerable improvement in forecast performance is observed by implementing the framework (Fig. 4, Fig. S2, Fig. S3). For the Rabi season, with the ECMWF model it is observed that for the Z2 region, the framework performance is lower than NWP forecasts for all the five ML/DL algorithms. Therefore the framework is not recommended for the Rabi season in the zone 2 using the ECMWF model. None of the ML/DL models were found to be particularly superior for a season and performance varied across the seasons.

4.4. Comparison of the proposed framework with persistence based approach

Here, we present the results of our proposed (NWP-based) ML/DL framework against the results from the persistence based approach. We applied the persistence-based approach across the entire Indian region to demonstrate the benefits of incorporating NWP forecasts within the proposed framework. The XGB model was used for this approach, as it provided comparable performance to other methods while requiring significantly lower computational resources (Table 2).

Figure 5 and S4 presents the NRMSE and R^2 of the NWP-based forecasts, the persistence approach (i.e., the proposed framework without NWP inputs), and the proposed ML/DL framework across different zones at lead time 1 and 7 respectively. The results indicate that, in terms of NRMSE, for the ECMWF model, the proposed approach improves the forecasts for all zones except Z2 (Fig. 5) for shorter lead times. For the GEFS model, the performance is similar to that of the persistence-based approach. This can be attributed to the superior quality of ECMWF forecasts, which benefit from a larger ensemble size

and also due to better representation of processes within the models such as complex topography, convective systems and superior data assimilation at lead times. The R^2 results indicate that the proposed approach consistently achieves higher R^2 values than both the NWP models and the persistence-based approach. This improvement is observed across all zones, including Z2, for both NWP models. These findings highlight that overall the proposed approach provides better predictive skill compared to the persistence-based method, especially when using the ECMWF model except for Z2 region. The forecast skill of the proposed approach is relatively limited in Z2 (as indicated by NRMSE), because the inclusion of NWP forecasts with comparatively lower skill, in this region adversely affects the training of the ML/DL models within the framework (Fig. 4, Fig. S2, Fig. 5 and Fig. S4).

At lead 7, the proposed approach consistently provides better forecasts than the persistence-based method for both the ECMWF and GEFS forecasts and even in Z2 region as well, as observed from the NRMSE and R^2 values (Fig. S4). These results highlight that at higher lead times the proposed framework can yield better predictive skill than persistence-based methods (Fig. 5, Fig. S4).

5. Discussions

This study focusses on proposing a data driven approach for forecasting ETo, by merging the NWP forecasts and persistence based approaches. We observe that the proposed approach provides enhanced forecasts compared to both the NWP based values, the BMA approach as well as persistence approaches.

A comparative analysis across other regions also show similar results. For example, Yang et al. (2019) and Medina et al. (2018) used NWP based ETo forecasts and found that the MAE values ranged from 0.6–1 mm/d in different regions in China and the United States (Medina et al., 2018; Yang et al., 2019), respectively, while the MAE values from the proposed study are better (range 0.1 to 0.6 mm/day). Other studies have also reported similar MAE ranges across different regions of the world (Fan et al., 2021; Pelosi et al., 2016). In India, the forecasts from ECMWF perform better than GEFS and we find that ETo forecast performance is particularly low in the Z2 region. This is due to the low temperature forecast skill prevalent in the region (Gerlitz et al., 2014; Saminathan et al., 2024; Saminathan and Mitra, 2024; Tiwari et al., 2016).

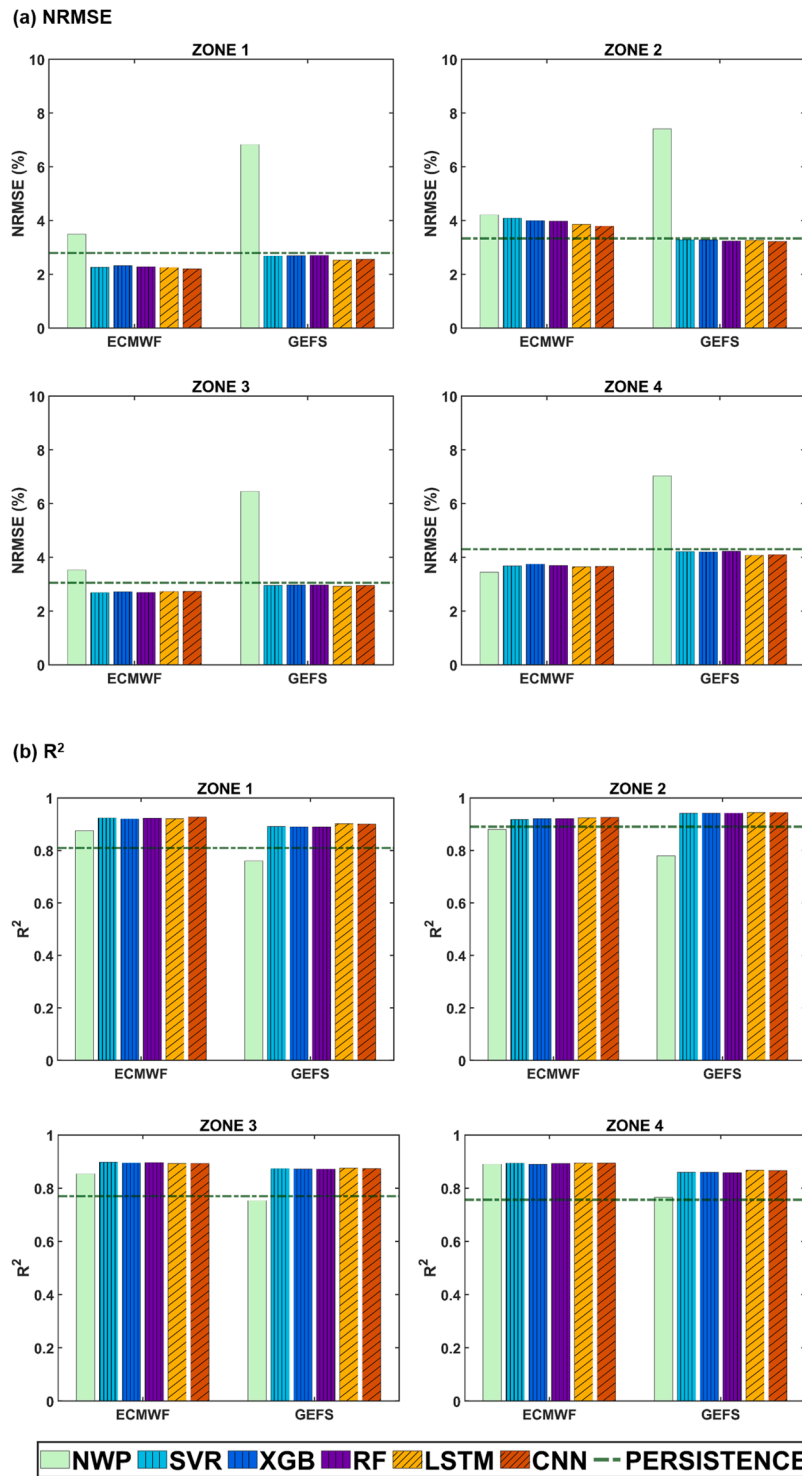


Fig. 5. Skill comparison of the proposed framework against NWP and persistence-based approach (using XGB only) for lead 1 day across different zones.

Countless studies have reported on forecasting ETo using persistence based approaches (Ahmadi et al., 2023; Ferreira and Da Cunha, 2020; Gong et al., 2021; Karbasi et al., 2022; Lucas et al., 2020; Saggi and Jain, 2019). A study by Ferreira and Da Cunha (2020) reported low R^2 values using CNN-LSTM models at 1 ($R^2 = 0.62$) and 7 ($R^2 = 0.50$) day lead in a region in Brazil (Ferreira and Da Cunha, 2020). ETo forecasting in southern Florida found LSTM to be the best model with good R^2 at 1 (0.915) and 7 (0.910) lead times (Granata and Di Nunno, 2021). Karbasi et al. (2022) found high RMSE (range 3–5 mm/d) values using XGB and LSTM models for forecasting ETo at 7 day lead (Karbasi et al., 2022).

However, in literature comparative evaluation of persistence based approaches with NWP based approaches are lacking. Here we show that over the Indian region, the proposed framework performs better than the persistence based approach, especially when using ECMWF model and across all regions except Z2 at shorter leads. At higher lead times, the proposed approach consistently outperforms the persistence approach across both models and all zones. This can be attributed to the diminishing influence of time-lagged ETo when using the persistence approach. A study by Zhang et al. (2023) used temperature forecasts from NWP models along with data driven approaches for forecasting

ETo in China (Zhang et al., 2023). Their results indicate that relying solely on temperature forecasts in ML/DL models may not yield superior forecasts, highlighting the need to use ETo forecasts from NWP models (Zhang et al., 2023).

The proposed approach was evaluated across different regions and agricultural seasons in India. The framework was able to provide skillful forecasts even in regions where the NWP forecast skill was relatively low (Zone 2). However, the level of skill achieved in these regions was lower compared to that obtained for the other zones, primarily due to the inherently lower performance of the NWP forecasts themselves. The poorest performance of the NWP models is observed in the zone 2, where both NRMSE and MAE values are notably higher (Fig. 4 and Fig. S2). In this region, the skill of ETo forecasts is relatively low compared to other zones, primarily due to the weak temperature forecast skill in the region which has already been explored in depth by Saminathan and Mitra (2025) (Saminathan and Mitra, 2025). Previous studies have consistently highlighted the limited accuracy of temperature forecasts in the Himalayan regions (Z2) (Durai and Bhadrwaj, 2014; Saminathan and Mitra, 2024; Tiwari et al., 2016). For instance, Tiwari et al. (2016) reported substantial winter temperature biases over the Himalayas, attributing them to the models inability to realistically capture the complex orography of the region due to limited spatial resolution. Similarly, Mishra (2015) and Gerlitz et al. (2014) noted that temperature simulations from climate models exhibit significant uncertainties in the Himalayan region, largely influenced by elevation variations and climate models resolution.

Further, forecasts were found to be better for the Rabi season compared to the Kharif season across the region. The Kharif season (June–September) coincides with the onset of the southwest monsoon and is typically marked by high humidity, elevated temperatures, and intense rainfall activity. These climatic conditions substantially increase the atmospheric demand, resulting in higher and more variable reference evapotranspiration (ETo) values. Such variability poses challenges for predictive models, as the fluctuations in meteorological parameters lead to greater uncertainty in ETo forecasting. In contrast, the Rabi season (October–March) corresponds to the winter cropping period, during which temperatures are relatively lower, rainfall is scarce, and atmospheric demand is subdued. Consequently, the ETo values during this season are generally lower and more stable, with reduced variability compared to the Kharif season. The relative stability of ETo during Rabi enhances the ability of ML and DL models to capture underlying patterns, thereby improving predictive performance. Consistent with these observations, Mandal and Chanda (2023) also reported that predictive accuracy tends to be higher under conditions of low and stable ETo (Rabi season), while performance declines with an increase in ETo magnitude and variability (Kharif season). This highlights the direct influence of seasonal climatic variability on the skill of ML/DL-based ETo forecasting approaches. Further, since ETo by definition is not dependent on the type of crops sown, therefore the accuracy of the values are not dependent on the crop themselves.

Overall, we do not find substantial difference between the different ML/DL models in forecasting ETo. However, few studies in literature have reported difference in performance of the ML models for forecasting ETo though those studies were solely based on persistence approaches (Karbasi et al., 2022; Mandal and Chanda, 2023; Reis et al., 2019; Saggi and Jain, 2019). Saggi and Jain (2019) compared four different ML/DL models such as RF, Gradient-Boosting Machine, Generalized Linear Model and Deep Learning-Multilayer Perceptrons and reported that the Deep Learning-Multilayer Perceptrons outperforms the other ML/DL models in forecasting ETo. Similarly Reis et al. (2019) compared three different ML approaches such multiple linear regression, extreme learning machine and artificial neural network, they found that multiple linear regression is best suited for ETo forecasting in Brazil. Furthermore, we also find that the performance of the proposed approach is considerably better than those of that of BMA a widely employed NWP bias correction technique. Many studies (Ahmed et al.,

2020; Barlow et al., 2018) have reported on the performance of such bias-correction approaches towards forecast enhancement, however we show that the proposed approach comprehensively outperforms the BMA approach. Overall, as all of the ML/DL models show similar performance within the proposed framework, the time taken for each model should therefore determine the choice of ML/DL models in the framework.

6. Conclusion

Short- to medium-range ETo forecasts are necessary for irrigation scheduling purposes and agricultural water management. In this study, we proposed an ETo forecasting framework integrating ML/DL algorithms with NWP forecasts and persistence based approaches. Further, we also compare different ML/DL algorithms within the proposed framework. The proposed framework was also evaluated across different climate zones and agricultural seasons over the Indian regions. The major conclusions from this study are as follows.

1. The proposed ML/DL-based framework was able to provide better ETo forecasts than the NWP models, whose forecasts exhibited high errors (NRMSE, MAE, RMSE) and are therefore unsuitable for direct end-user application.
2. Comparable performance was observed between the different ML/DL techniques (SVR, XGB, RF, LSTM and CNN), and no single ML/DL technique was found to outperform the other.
3. The XGB model demonstrated the lowest computational time among the evaluated models while achieving comparable predictive performance, making it the preferred approach for future ETo forecasting applications.
4. Performance assessment across major crop growing seasons indicated that the proposed framework provided better forecasts for the Rabi season compared to the Kharif, which is due to the low skill of the NWP based forecasts for the Kharif season compared to Rabi.
5. Comparison of the proposed ML/DL based framework against BMA and the persistence-based approach showed that the proposed framework performs better than both the approaches, especially for regions having higher NWP forecast skill. Further, the proposed framework is able to provide better results than persistence based approach and the BMA even at higher lead times.

The proposed framework based on integrating NWP-based forecasts and time lagged ETo observations within ML/DL algorithms can be used for operational purposes and will be helpful for the irrigation and farming community.

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Authors agreement

All authors have seen and approved the final version of the manuscript. All the authors warrant that the article is the original work, has not received prior publication, and is not under consideration for publication elsewhere.

CRediT authorship contribution statement

Sakila Saminathan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Subhasis Mitra:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Lahari Chowdary Talasila:** Formal analysis, Investigation, Resources,

Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.agrformet.2026.111414](https://doi.org/10.1016/j.agrformet.2026.111414).

Data availability

Data will be made available on request.

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